



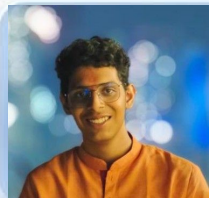
BENNETT
UNIVERSITY
TIMES OF INDIA GROUP

CAMPUS CONNECT: Know Your Bennett.

Presented by **SqlChamps.**
Kshitij Dhanetwal, Divyapunj, Ritesh Gocher.



MEET THE TEAM



Kshitij Dhanetwal

Role Title Placeholder



Divyapunj

Role Title Placeholder



Ritesh Gocher

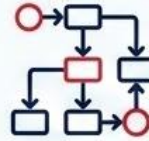
Role Title Placeholder



Agenda



**1. Problem &
Objective**



**2. System
Architecture**



**3. Tech Stack &
Implementation**



**4. Testing &
Validation**



**5. Output
Snapshots**



**6. Conclusion &
Future Scope**

Abstract & Executive Summary

Campus activities are often hindered by fragmented communication across WhatsApp groups and emails, resulting in low student engagement and disorganized record-keeping. CampusConnect is a centralized, web-based event management system engineered using Java and Spring Boot. By providing a unified platform for event discovery, seamless student registration, and efficient administrative control, this project eliminates information silos, streamlines campus logistics, and significantly boosts overall student participation.



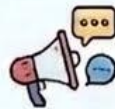


Introduction

2.1 Background of Project



College events include workshops, seminars, competitions and fests.



Information is currently shared through WhatsApp groups, emails, and social media.



These communication channels are **scattered and unorganized**.



Students may miss important **announcements or opportunities**.



A centralized event management platform is required.





2.2 Problem Statement

- **Information Silos:** Key updates hidden across multiple apps.
- **Notification Overload:** Critical alerts lost in the noise.
- **Missed Deadlines & Events:** Opportunities overlooked due to disorganization.
- **Student Frustration:** High stress from managing fragmented information.



2.3 Object & Scope of Project

Objective:



Objective:



Develop a centralized platform for managing event information.



Provide a simple registration system for students.



Enable administrators to efficiently create and update event details.

Scope:



Scope:



Students can explore event details and register directly.

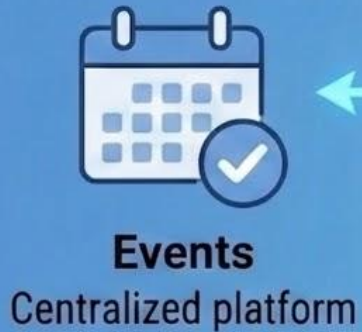


Administrators can manage information within a structured MySQL database.

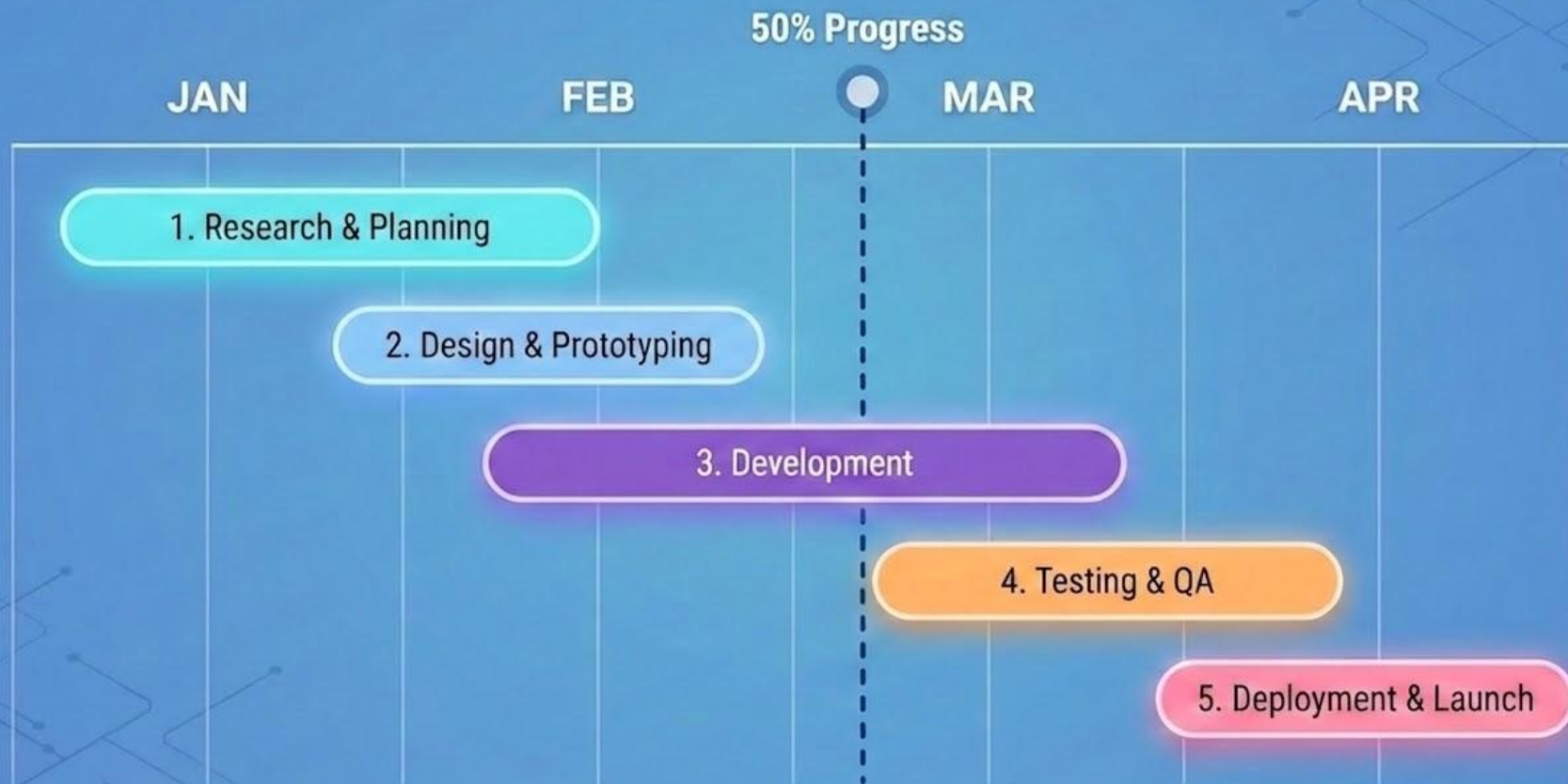


Improves overall campus communication and accessibility.

2.4 THE SOLUTION (Objectives & Scope)



2.5 Project Timeline (Gantt Chart)



Literature Review

3.1 Overview of existing system



- Event information is mainly shared manually through emails, WhatsApp, and social media.
- Students rely on multiple channels to receive updates.
- There is no centralized platform to view all upcoming events in one place.
- Students frequently miss important announcements and registration deadlines.



3.1 Overview of Existing System(Comparison)

Feature	Standard Manual (WhatsApp/Forms)	Top-Tier Fest Portals (e.g., IITs/VIT)	CampusConnect
Centralized Daily Hub	✗ No	⚠ Temporary (Fests Only)	✓ Yes (Year-round)
Student Club Integration	✗ Scattered	⚠ Limited to Mega-Events	✓ Fully Integrated
Automated Record Keeping	✗ Manual Export	✓ Yes	✓ Automated via MySQL
Prevents Notification Spam	✗ No (Message Overload)	✓ Yes	✓ Yes

While institutes like IITs and VIT have strong event cultures, their custom portals are often built only for annual mega-fests. Everyday club activities still suffer from decentralized chaos.

3.2 Limitations of Existing Models



Fragmented Daily Life

Scattered WhatsApp groups for daily events cause message fatigue (e.g., Ashoka, NIIT).



Temporary Infrastructure

Custom portals for fests are temporary, failing year-round needs (e.g., IIT Techfest).



Registration Bottlenecks

External links (Google Forms) create disjointed user experience and hinder participation tracking.



Lack of Centralized Archives

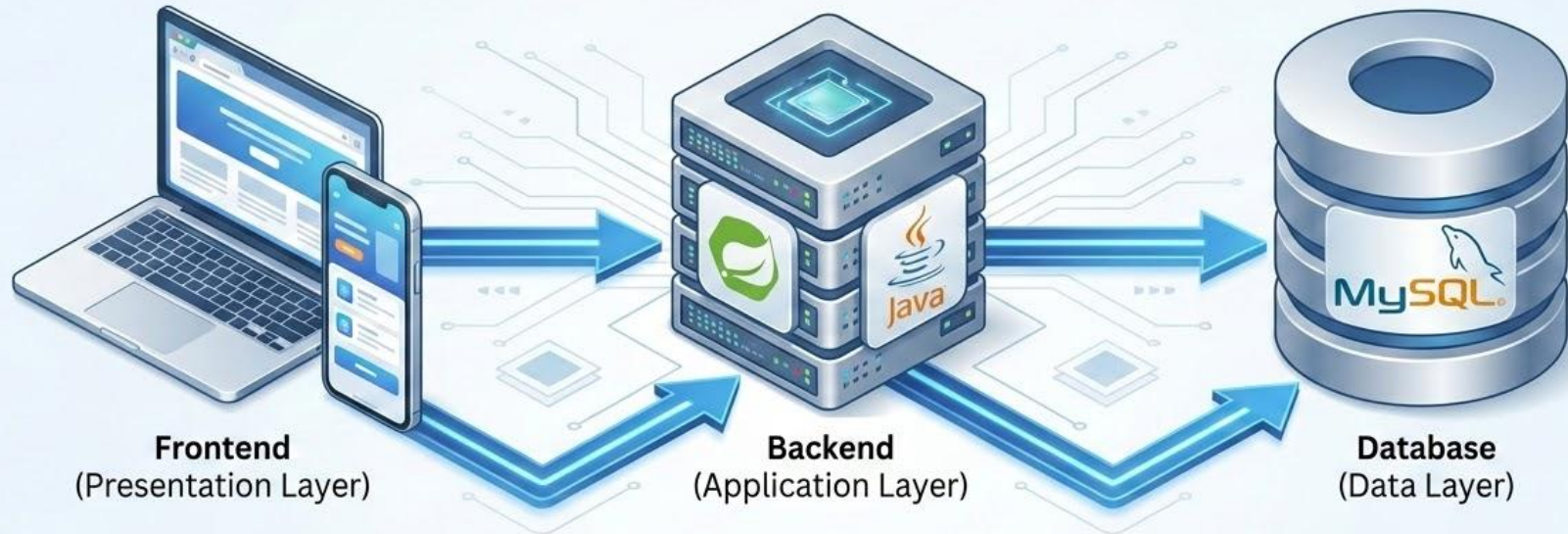
No single database of past events, making long-term engagement measurement difficult.

3.3 Proposed System: CampusConnect

-  A permanent, centralized web platform engineered with Spring Boot and MySQL.
-  Brings the high-tier digital organization of "mega-fests" to everyday campus life and departmental workshops.
-  Provides a unified dashboard for students to discover and register for events with a single click.
-  Equips administrators with robust tools to track registrations, manage schedules, and maintain permanent event archives seamlessly.



4.1 System Overview



- **Architecture:** CampusConnect utilizes a robust 3-tier Client-Server architecture to ensure high performance and scalability.
- **Frontend (Presentation Layer):** Responsive web interfaces built with Thymeleaf, HTML5, and Bootstrap for seamless student interaction.
- **Backend (Application Layer):** Powered by Java and Spring Boot to securely handle business logic, event scheduling, and user authentication.
- **Database (Data Layer):** A centralized MySQL database ensuring data integrity for all event records and student registrations.

4.2 Advantages of Proposed System



Centralized Information Hub

Eliminates scattered WhatsApp messages and emails by putting all campus events on one unified platform.



Streamlined Registration

Students can view details and register for events with a single click, significantly boosting participation.



Automated Record Keeping

Replaces manual Google Forms and Excel sheets with a robust MySQL database for instant tracking.



Enhanced Administrative Control

Gives organizers a dedicated dashboard to effortlessly create, update, and manage event schedules in real-time.

System Design (UML Diagrams)

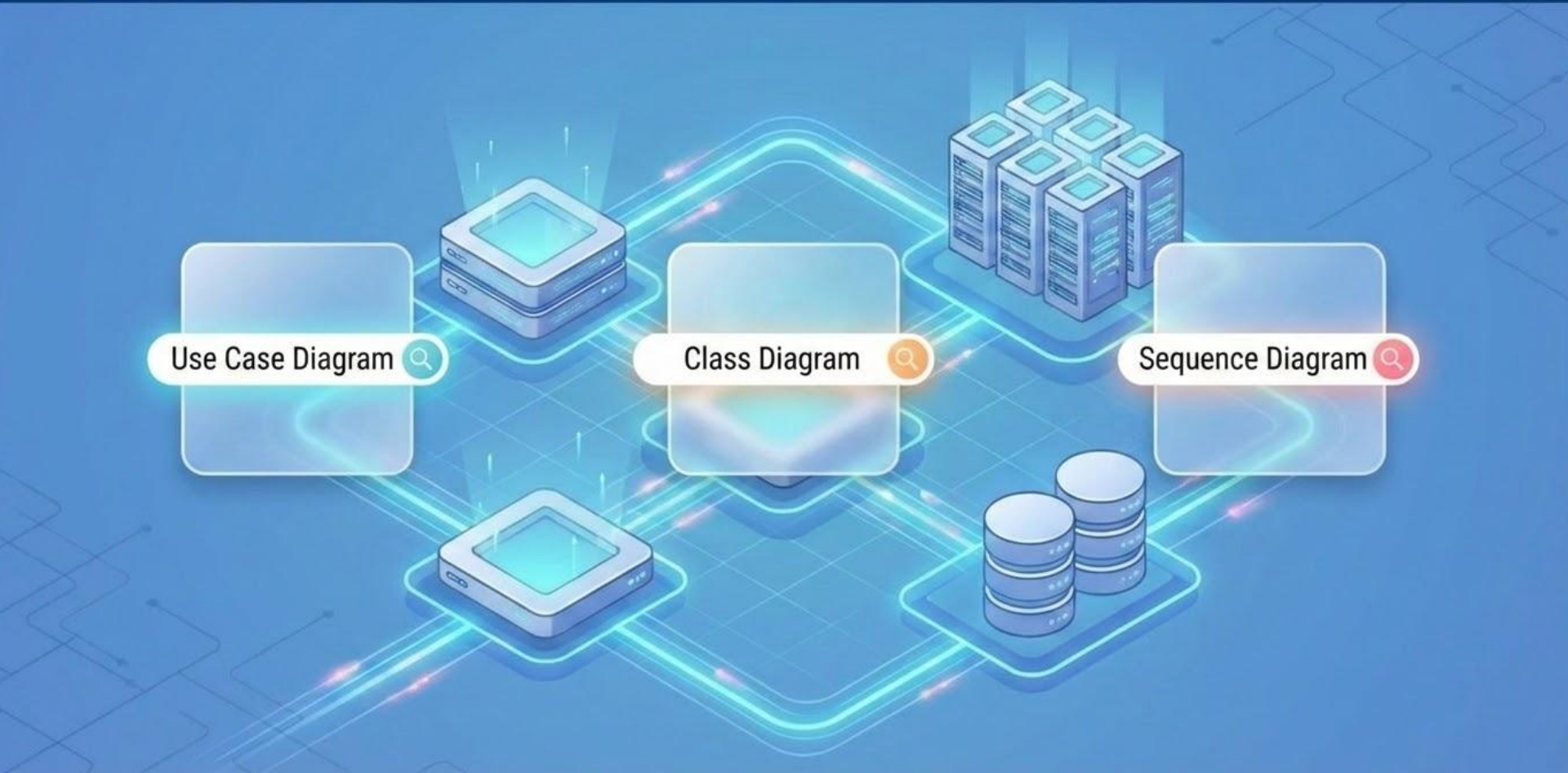
Use Case Diagram



Class Diagram

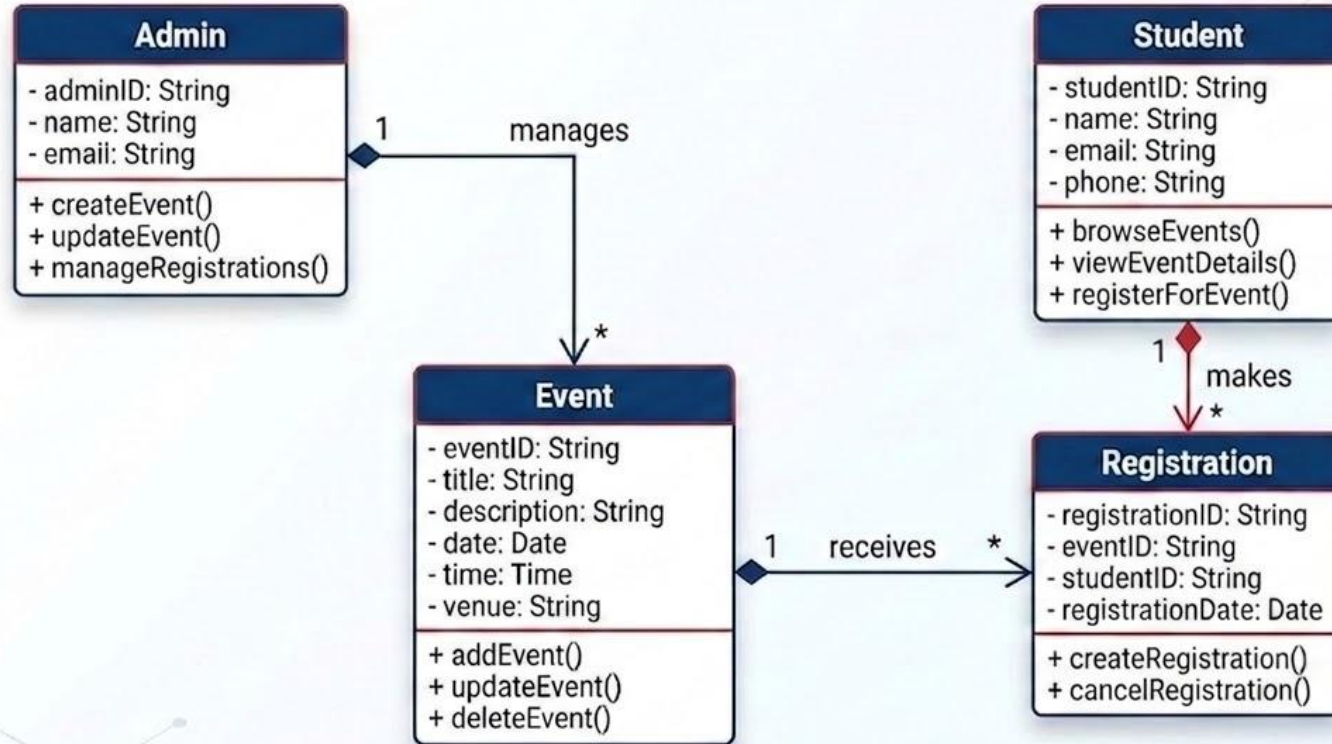


Sequence Diagram



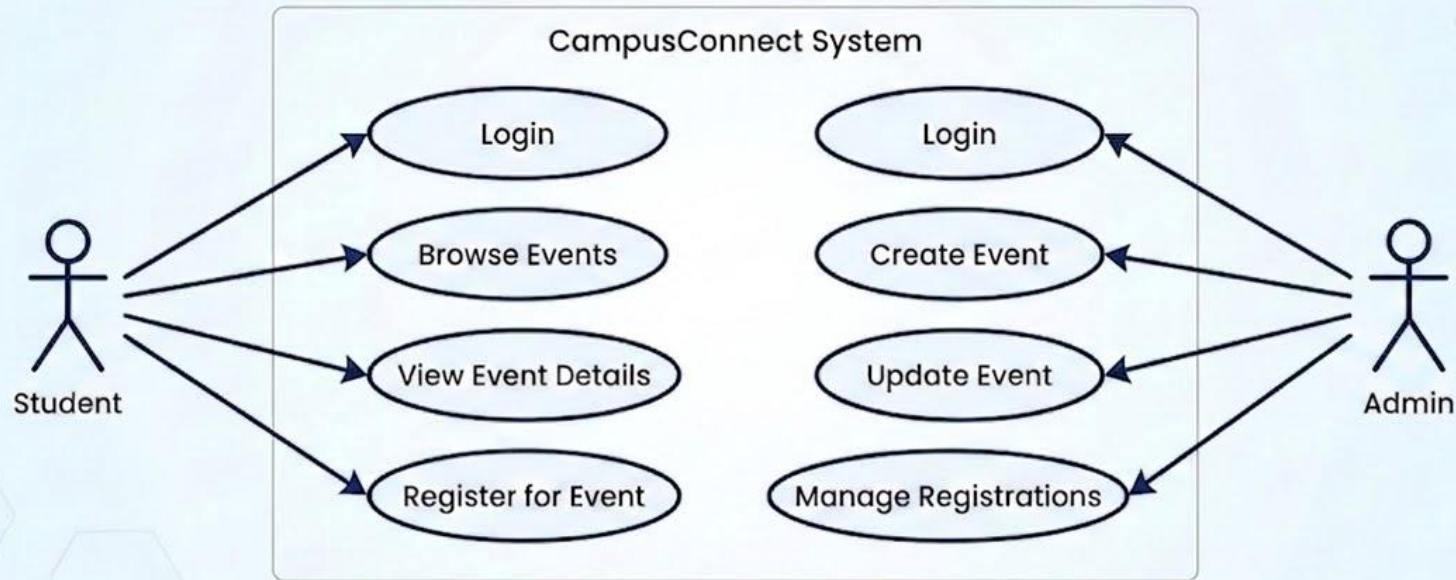


5.1 Class Diagram



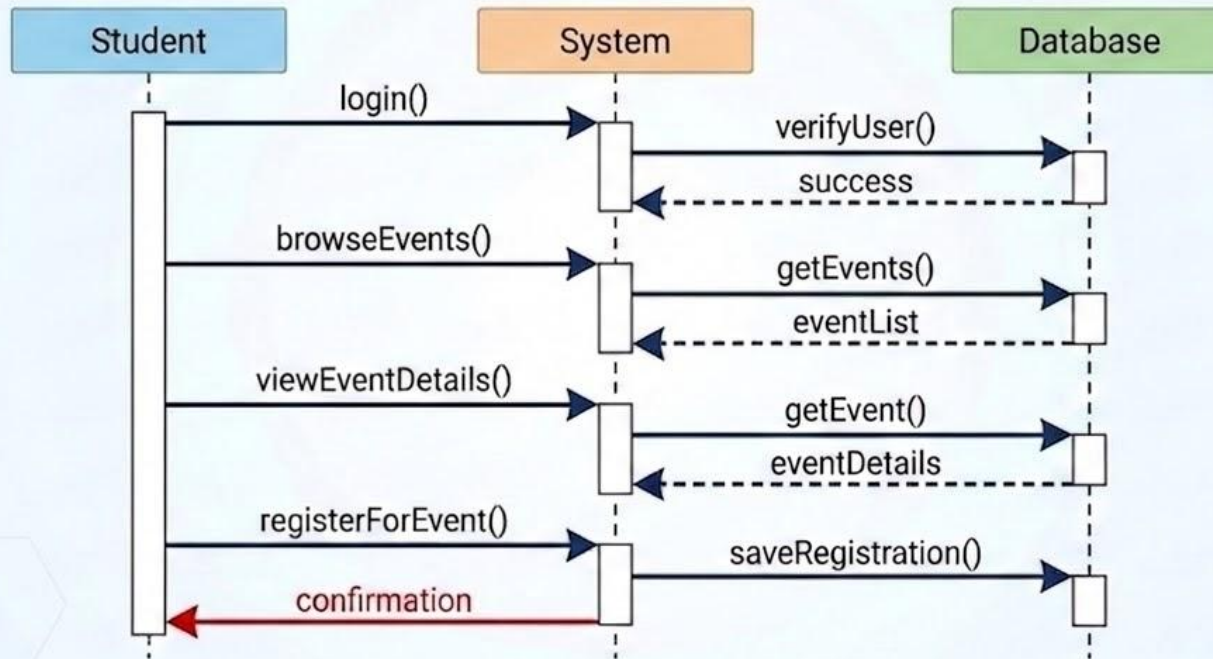
The Class Diagram illustrates the core entities of the CampusConnect system: Admins who manage events, Students who browse them, the Events themselves, and the Registration records linking students to activities.

5.2 Use Case Diagram



The Use Case Diagram illustrates the primary interactions within the CampusConnect system. Students interact with the platform to browse and register for events, while **Administrators** are responsible for creating events and managing overall registrations.

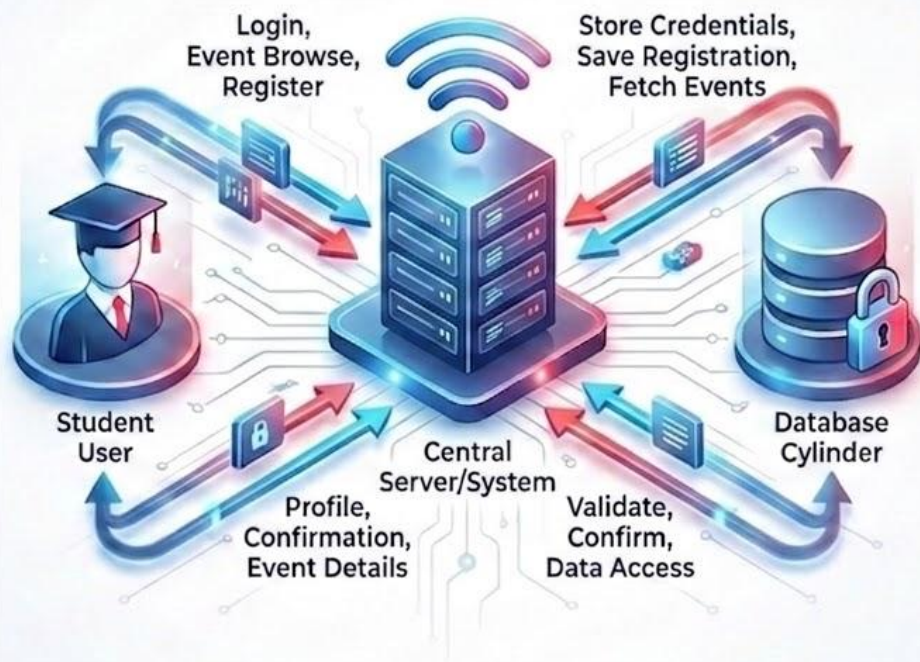
5.3 Sequence Diagram



The Sequence Diagram details the chronological flow of a student interacting with the CampusConnect platform. It highlights the step-by-step process of logging in, fetching event data from the backend, and securely saving a new event registration into the database.

5.4 Data Flow Diagrams (DFD)

- **Level 0 (Context Diagram):** Illustrates the macro-level interactions between the CampusConnect System and external entities (Students & Admins).
- **Level 1 Diagram:** Showcases the exact flow of data through:
 - User Authentication
 - Event Management & Updates
 - Event Registration
- **Data Stores:** Highlights, how information routes securely to the User, Event, and Registration databases.





5.5 Database Design & ER Architecture

- **Relational Schema:** Built on MySQL to ensure robust data integrity, ACID compliance, and seamless query performance.
- **Core Entities:**
 - **Users Table:** Securely stores credentials and roles (Student/Admin).
 - **Events Table:** Houses event schedules, venue locations, and detailed descriptions.
 - **Registrations Table:** The junction table tracking which student is attending which specific event.
- **Data Relationships:** Utilizes strict One-to-Many (1:N) relationships and foreign key constraints, ensuring accurate cascading if an event is modified or deleted.

Implementation of Modules

6.1 Module Description



6.2 Module-wise Implementation Details



Security & Auth

- Implemented using Spring Security.
- Configured role-based access control to distinctively route Students and Administrators upon login.



Backend Logic

- Built utilizing standard Spring Boot Controllers and Services.
- Handles dynamic routing, event creation logic, and mapping user requests to the database.



Data Persistence

- Implemented via Spring Data JPA & Hibernate.
- Uses Entity classes mapped directly to MySQL tables to perform seamless CRUD operations without raw SQL.

6.3 Tools & Technologies Used

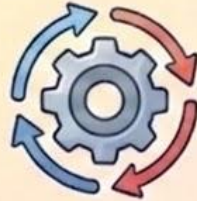


7.1 Working of Each Module



Security & Authentication

Validates users, roles, & routes securely.



Admin Operations

Admins manage & publish campus events seamlessly.



Student Registration

Students browse & sign up for active events.



Data Persistence (JPA)

Efficient, secure database interactions without raw SQL.

7.2 Output Screenshots: Authentication

- Secure Login & Registration portals.
- Role-based redirection (routes Admins and Students to their dashboards).
- Password encryption handling via Spring Security.



7.2 Output Screenshots: Admin Control Panel

- Dedicated workspace for university organizers.
- Real-time event creation, updating, and deletion.
- Seamless management of campus announcements and monitoring student registrations.



7.2 Output Screenshots: Student Portal

- Centralized feed of all upcoming university events.
- Dynamic filtering and structured layout for easy discovery.
- Direct access to official announcements without the clutter of WhatsApp groups.



7.2 Output Screenshots: Event Registration

- Seamless, one-click registration process for students.
- Direct data binding to the MySQL backend.
- Instant confirmation and schedule updates.





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8.1 Output Analysis



Zero Errors

- All core modules execute flawlessly.



Data Integrity

- MySQL securely processes concurrent registrations.



Responsive UX

- Dynamic UI with instant visual feedback.



High Performance

- Spring Boot MVC architecture ensures minimal load times.

8.2 Performance Evaluation



Metric 1: Response Time & Latency

Avg. load < 1.5s. Instant UI via efficient Thymeleaf.



Metric 2: Concurrency Handling

Spring Boot handles simultaneous registrations, no DB locking.



Metric 3: Resource Utilization

Optimized MySQL/JPA reduces memory. High peak stability.



9.1 Conclusion



Problem Solved



- Centralizes fragmented campus communication into one unified platform.
- Replaces cluttered WhatsApp groups and scattered emails.



Operational Success

- Streamlined event management for organizers.



- Automated, secure backend via Spring Boot & MySQL.

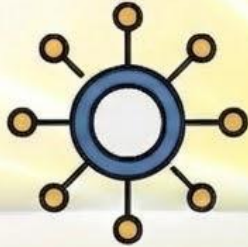


Overall Impact

- Fundamentally boosts student participation.
- Creates a more vibrant, connected, and engaged Bennett University campus.



9.1 Conclusion



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9.2 Future Scope



Integration with **mobile** applications (Android/iOS)



Addition of **AI-based** event recommendations



Implementation of **chat or** discussion forums



Enhanced **security** features (OTP, biometric login)



Integration with **college** ERP systems



Support for **online event** hosting (live sessions/webinars)

Thank You! :)

Any Questions?

